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10 Data-Driven Artificial Intelligence (AI) Models in the Workforce Development Planning

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and Alex Khang*

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10.1 INTRODUCTION

There has been a significant advancement in corporate management thanks to the introduction of artificial intelligence (AI), which will have a significant effect on the way employees perform their jobs, particularly in the areas of human resources (HR) (Rodríguez et al., 2022). AI-powered technologies affect HR administration in distinctive ways.

Big data and data analytics can be used to tailor training and development programs to each employee, as one example, by shedding light on HR practices in real time. The worker's prior practices might provide a clue as to where to find these strategies (Gupta et al., 2023).

The term “artificial intelligence” refers to the use of technology to perform a task that typically requires a certain level of intelligence to complete successfully. Basically, it's a device that can be programmed to carry out activities previously thought to require human intervention (Khang, Rani & Sivaraman, 2022).

The application of AI in a way that is both practical and effective has led to an improvement in the accomplishment of the work tasks associated with human resource management (Eubanks, 2022).

Employment, assessment and performance measurement, HR planning, training needs of employees, job evaluation, and even forecasting of the labor market and its requirements and indicators have all experienced improvements as a result of these developments (Di Vaio et al., 2020).

The use of algorithmic data mining (DM) has soared over the past decade as a result of the availability of vast datasets, the increased processing power of computers, and cloud storage. AI is present in virtually every aspect of modern life (Rana et al., 2021).

Combining low-risk usage scenarios with practically infinite access to behavioral data gathered from social media, sensors, and web-based platforms has boosted machine learning (ML) models’ performance (Khanh & Khang, 2021). However, the achievement of this success has resulted in a decreased awareness of concerns, such as the ability to validate and verify claims and to explain them (Ahern et al., 2019). Failure is not a catastrophic event in low-risk scenarios, and profiling models get better as a result of the lessons they learn from their own errors. Recommendation systems ask users to rectify mismatched profiles.

Data-driven AI built on ML techniques is used to assist DM in justice, law enforcement, medical services, education, and public service (Bierbaum, Vincent & Katz, 2022). Prediction and DM algorithms now affect people’s lives. Early development must foresee ethical and epistemological issues.

The three basic categories of workforce planning strategies that can be adjusted to each organization’s particular goals and overall strategy are strategic, operational, and tactical. Each technique investigates its own particular goal, time, and amount of planning. Three workforce planning categories are illustrated in (Hurley & Hutchinson, 2022) [Figure 10.1](#) (Dunn, 2018).

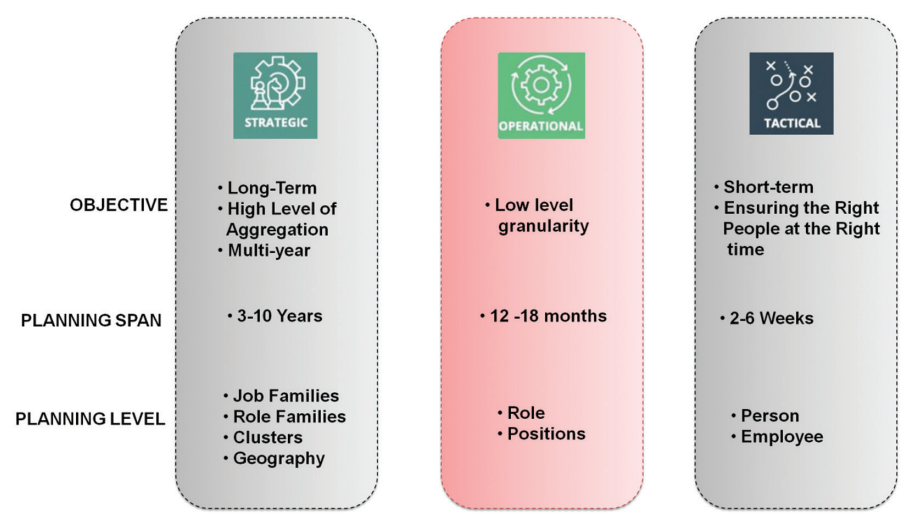


FIGURE 10.1 Three workforce planning categories.

Consideration and computational evaluation of value-based design needs should also be included in the discussion (Frady, 2020). To prevent unintended value outcomes brought on by obtuse “self-learning algorithms,” you should instruct a “tax fraud detection algorithm” to optimize itself for efficiency and fairness.

The computational complexity prohibits the use of a one-step approach. By setting out to decrease mistakes without upsetting some preconceived notion of justice, we impose a restriction on learning that comes at the expense of precision (Khang et al., 2023a).

Although the tax fraud models we generate using our optimization method are fairer, they are also less precise. Since compromising precision for moral algorithms will reduce efficiency, we need to reach consensus on the appropriate threshold for such concessions (Holland, 2019).

Here, we need to strike a balance between fairness and precision to reduce the number of false positives and missed tax frauds. It’s possible to complete the work computationally, but the steps are trickier than they seem. Still, value hierarchies, which translate values and norms into design requirements, can facilitate interdisciplinary conversations about integrating ethical standards in AI design (Barracough et al., 2021).

The suggested AI for social good–value-sensitive design (AI4SG-VSD) methodologies shed light on AI-specific value issues throughout system development. Taking an AI4SG-VSD approach necessitates a well-established multidisciplinary framework (Prikshat, Dhakal & Nankervis, 2020).

The authors’ ignoring of these difficulties is troubling, especially given their abstract, highly conceptual prototyping technique. Despite the necessity of multidisciplinary collaboration, they advise a traditional approach to value-sensitive design (VSD) (Hoge et al., 2019).

10.2 PROBLEM AND MOTIVATION

The capacity to adapt to changes in a timely and efficient manner is referred to as enterprise agility. This skill is a significant obstacle for modern-day commercial organizations. Enterprises are dependent on the ever-changing nature of the business environment in which they operate and do business.

As a result, they are susceptible to unanticipated changes and uncertainties, both of which almost always have unfavorable outcomes and have the potential to bring about unfavorable conditions such as financial loss, bankruptcy, and the loss of a competitive advantage for the enterprise that is serving as the host.

In addition, agility improves an organization’s capacity to optimize profits, maintain and grow its client base, and expand its market share. In spite of these advantages, business agility remains a formidable obstacle, an unreachable goal, a tough objective to realize, and a big source of stress for corporate leaders (Babasaheb et al., 2023a).

The present academic effort towards actualizing agility, no one understands how to attain agility to this day, whereas businesses are still looking for more effective strategies to that increase agility (Mendez, 2021). Therefore, these techniques can help support the capturing and management of data relating to manage the changes. In order to facilitate the effective documentation, storage, and management of changes, you need the necessary action plans, rules, and procedures needed to anticipate or respond to these changes.

In addition to this, the use of such methods might make it easier to reuse tactics and ideas. In other words, the strategies and methods that were used during earlier change-management activities may be retrieved and applied to assist future choices pertaining to change management (Babasaheb et al., 2023b).

In addition, strategies for data modeling may improve the efficiency with which corporate stakeholders share information, a crucial factor in determining the degree to which an organization can be agile. Equally, a conceptual data model may be utilized to construct a database system that will assist in the process of change management (Ozkan-Ozen & Kazancoglu, 2021).

In addition to its capacity to ease the exchange of information, the visualization of data, and the administration of information, a database gives the functionality to conduct searches, create forms, and generate reports. This can be done in order to determine which changes should be addressed first. It is possible to cut down on the amount of time, money, and effort required making choices on change management in this manner, while simultaneously increasing the efficiency of the use of corporate resources.

Despite these potential advantages, data modeling approaches in relation to corporate agility and change management are seldom included, if at all, in the current change management frameworks and methodologies.

10.3 SELECTED FAMILY MEMBERS IN THE AI VALUE SENSITIVE DESIGN FIELD ARE INTRODUCED

The “design turn” in IT ethics during the 1980s and 1990s advocated for a proactive approach to IT system development by placing an emphasis on “front-loaden” ethics rather than post hoc evaluations of IT systems.

Friedman and Khan (2007) introduced value-centered design (VSD) as an ethically-focused alternative to related user-centered and participatory design methodologies (CSCW) in “Human Computer Interaction (HCI) and Computer Supported Cooperative Work pioneers.” The goals of VSD were admirable; it hoped to get social scientists and humanists working side-by-side with IT pros to create something truly innovative in design.

To describe the part that the humanities and social sciences play in the creation of IT systems, Nissenbaum (2001) invented the phrase “engineering activism.”

Humanists and social scientists can no longer bracket technical details—leaving them to someone else—as they focus on the social effects of technology. Fastidious attention to the before-and-after picture, however richly painted, is not enough. Sometimes a fine-grained understanding of systems—even down to gritty details of architecture, algorithm, code, and possibly the underlying physical characteristics—plays an essential part in describing and explaining the social, ethical, and political dimensions of new information technologies.

Conceptual, empirical, and technological research are all components of the VSD technique. Usually, the conceptual level is where one starts. Philosophical investigations help find important values and identify system stakeholders.

In practice, designers take into account the values and opinions of both direct and indirect stakeholders. Technical investigation translates values into technical requirements. The tripartite iterative VSD approach hasn't produced a research agenda in recent years.

VSD stresses philosophical appraisal of existing technologies and techniques or conceptual exploration of values with stakeholders. VSD collaborates less with developers to incorporate values in IT systems. Technical problems, such as cooperation between social scientists, humanists, and system developers, are disregarded.

As highlighted by Nissenbaum 20 years ago, in order to embed values in AI systems, gritty detail must be paid attention to. Encourage interdisciplinary research and respect for the perspectives of others. Contextual ethical issues must be taken into consideration by AI developers. It is necessary for sociologists and humanists to grapple more vigorously with technological solutions and make sure the computer community is aware of operational concerns.

Sociologists and humanists run the risk of turning discussions about technology and ethics into philosophical debates if they stay out of the technological details. Converting values into technical design requirements is a necessary step in facilitating stakeholder-driven value elicitation (Hussain, Sivakumar & Khang, 2022).

The worst-case scenario for ethical AI design is that it might be seen as "silo-derived stakeholder-informed moral philosophical counsel." If social scientists and humanists commit to interdisciplinary collaboration, they can clarify values in a way that informs technical design.

Humanists may avoid incorrect, mythically exaggerated views about AI's issues if they work with AI developers. The misconception is that ML algorithms are self-learning black boxes. ML algorithms are hand-engineered and well-understood.

Rupert Morrison discusses the "Organization System" in our training course titled "Data-Driven Organization Design and Workforce Planning." This system can assist organizations in gaining a better understanding of the data that is readily available within a company as well as how the various aspects of work and workforce can be integrated together.

In addition to this, it demonstrates how workforce planning and organization design are essentially two sides of the same coin. Within the framework of the Organization System, there is a definite divide that exists between positions and roles.

In the process of designing roles such as human resources business partner (HRBP), organizations plan for positions (such as HRBP for finance, HRBP for marketing, etc.), which are subsequently filled by individuals. It is helpful to think of the work itself in terms of its goals, actions, and the knowledge and abilities required doing the work.

Each of these components ought to be comprehended through the utilization of data, whether it be inside or external to the organization. For instance, one can collect information regarding one's skills from a variety of venues. Workforce planning based on a data-driven AI model is depicted in [Figure 10.2](#).

These algorithms build non-explainable, multilayered neural networks. Simple algorithms, like optimization algorithms, operating on complicated data yield complex and sometimes unintelligible models. Elish and Boyd (2017) use Cambridge Analytica's technical advantages to illustrate mythical hype.

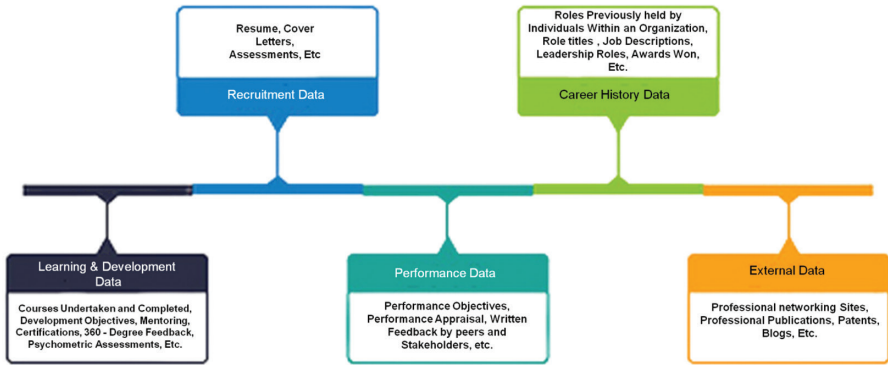


FIGURE 10.2 Workforce planning based on a data-driven AI model.

Elish and Boyd (2017) contend that when ML specialists entered the discussion and discussed the current state of the art in the field, the exaggerated idea that Cambridge Analytica may be held accountable for influencing elections was significantly dampened.

10.4 A CLOSER EXAMINE OF A FEW AI4SG-VSD TECHNIQUES

We describe two archetypal approaches to *ethical AI design*—the *Sienna (2021) project* and the *AI4SG-VSD method*—against the backdrop of AI misunderstandings, mostly as a result of a lack of understanding of what is computationally feasible. As part of its *VSD to establish a framework for AI Ethics* by design, Sienna considers ethical considerations at every stage of the AI life cycle, from inception to implementation.

The authors provide a five-tiered model that progresses from generalized value levels to concrete “ethical requisites” for design. This allows AI designers to proactively synthesize values without compromising the usefulness of the system. They also offer advice on how to prepare for ethical difficulties, including how to record risk mitigation measures by outlining the criteria used to clean data and eliminate bias from training datasets (Khang et al., 2023b).

Also included is advice on planning future steps after implementing a system. Insights from the five-layer model are used to develop instructions and tasks for AI developers in the “specification of objectives, specification of requirements, high-level design, data collection and preparation, detailed design and development, and test” phases of a generic system development process (Rani et al., 2023).

Here, we describe these stages with an emphasis on data-related tasks. From the start, stakeholders’ input must inform design choices and goal-setting. This guideline emphasizes the significance of developing mutual understanding between stakeholders and developers before data exploration. The needs phase presents data tools (Bhambri et al., 2022).



FIGURE 10.3 Importance of workforce planning.

Datasheets describe dataset features and recommended usage scenarios to assist analysts for reflecting reflect on the several phases of a dataset’s lifecycle: data understanding, data preparation (collecting, cleaning, labelling, preprocessing), modelling, evaluation, deployment, and maintenance.

Data statements are “a characterization of a dataset that gives context to help developers and consumers understand how experimental results might generalize, how software might be deployed, and what biases might be reflected in systems built on the program” (Jain et al., 2023).

Documentation tools can aid data governance and auditing. Importance of workforce planning is indicated in [Figure 10.3](#).

Model cards “tag” ML models by defining key application attributes to prevent models from being transferred indiscriminately. The high-level design phase uses these technologies to assure system transparency and ethics.

A development process may be governed by organizational procedures, such as evaluation, data security evaluations, testing regimes, paperwork of data activities, and organizational procedures resolving ethical governance authority and AI developer conflicts (Gupta et al., 2023). These non-technical requirements are required to make these processes possible. Data collection and preparation are also crucial for fairness and accuracy (Khang et al., 2022a).

“Data may never be assumed to be reliable, representative, or neutral; it must be shown,” the authors caution. Data privacy, data transparency, and bias reduction should all be taken into account while selecting data and training datasets. The design is checked against the overall ethical guidelines during the extensive design phase.

During testing and assessment, “a testing regime is developed using the project’s ethical requisites document.” Concerned parties should provide input on whether or not the system complies with ethical standards. In conclusion, ethical guidelines for execution, mobilization, and use are debated, together with suggestions for additional work, including risk analyses and techniques to ensure ethical manner secure

integration and to predict possible ethically questionable adjustments in the embedded “ethical attribute of the system.” Since the platform may evolve during deployment if new datasets are uploaded, post-deployment maintenance is essential.

The AI ethics by design methodology incorporates deployment and organizational procedures affected by workflow reorganization as well as broader organizational cultural problems. Prikshat, Dhakal & Nankervis, 2020 call VSD “the widely preferred technique for AI development,” yet their AI ethics by design method collapses VSD into an “add-on tool” to system development approaches. The authors base their AI4SG-VSD technique on two interrelated AI issues that VSD overlooks.

First, AI systems may gain information in opaque or unintelligible ways, challenging DM legitimacy and responsibility.

Second, AI systems may go astray and learn in ways that transgress or override system values. Subtle, unexpected, and opaque biased learning processes with negative effects may lead to moral transgression. It’s crucial not only to anticipate ethical concerns to avoid moral wrong (VSD’s major thesis), but also to ensure that AI systems are useful and promote social good values.

The authors define “sustainable development goals (SDG)” as internationally shared social aims. They expand VSD to integrate AI system deployment, much like the Sienna Project. First, they suggest using “a collection of AI-specific design principles” instead of the VSD heuristic value list.

Here, they emphasize the EU High-Level Expert Group’s principles of respect for individual autonomy, damage minimization, justice, and explicability. Bottom-up user-value elicitation is possible. Include a general list from *an AI-specific entity* to avoid omitting well-known AI-ethical dilemmas. AI4SG is based on seven ethical principles rooted in the above values.

Falsifiability and incremental deployment necessitate formal verification of system states. Self-learning systems can change their models to reflect dynamic inputs. It’s crucial to determine if the system meets restrictions or if the AI-system functions as expected. Safety-critical sectors require falsifiable assumptions for deployment.

“Predictor manipulation safeguards” limit algorithmic gaming and non-causal pattern overestimation in data analytics. “Receiver-contextualized intervention” emphasizes balancing human autonomy and decision authority with technology.

Profiling technologies can increase data-driven contextualization as long as they don’t violate autonomy. Users must be able to disregard some interventions but accept others afterwards. “Receiver-contextualized explanation and transparent purposes” promote avoiding opaque actions and emphasizing explainable interfaces for providing domain-relevant explanations (Khang, Rani & Shah 2024).

In light of VSD’s shortcomings and the addition of AI-specific characteristics, the authors suggest the four-phase AI4SG-VSD technique. In this study, direct and indirect stakeholders and socio-technical difficulties are highlighted. In the second, non-empirical phase, SDG, AI, and contextually relevant values are identified. Based on the first two steps, the third formulates design needs.

The authors recommend using a value hierarchy tool to illustrate “potential design paths” to transform abstract values into design needs. “Nonmaleficence” translates to “privacy protection and data subject consent” farther down the hierarchy. At the lowest level, design standards such as “clear conditions of usage” are given.

Value elicitation lets designers design for numerous values through visualization. System design and deployment may be revised after prototype testing. Prototypes and mockups lack precision for AI ethical system design and ML deployment maintenance. These technologies won't help you overcome AI design issues and give clarity.

It is convenient to test new ideas in small scale via prototypes. However, regularly relying on a prototyping environment may be an indicator that the full-scale system is brittle, difficult to change, or could benefit from improved abstractions and interfaces. Maintaining a prototyping environment carries its own cost, and there is a significant danger that time pressure may encourage a prototyping system to be used as a production solution. Additionally, results found at small scale rarely reflect the reality at full scale.

However, the inter-disciplinary nature of the design process, which is essential to the success of AI4SG, is not explicitly addressed in the scope of the AI4SG-VSD approach. Building an AI without a bridge builder seems unlikely. Fairness "is not just an abstract restricted optimization problem," hence value elicitation must be philosophical. It's a complicated, contextual, socio-technical challenge that must be tackled.

Consideration and computational evaluation of value-based design needs should also be included in the discussion. To prevent unintended value outcomes brought on by obtuse self-learning algorithms, you should instruct a "tax fraud detection algorithm" to optimize itself for efficiency and fairness.

The computational complexity prohibits the use of a one-step approach. By setting out to decrease mistakes without upsetting some preconceived notion of justice, we impose a restriction on learning that comes at the expense of precision. Although the "tax fraud models" we generate using our optimization method are fairer, they are also less precise. Since compromising precision for moral algorithms will reduce efficiency, we need to reach consensus on the appropriate threshold for such concessions.

Here, we need to strike a balance between fairness and precision to reduce the number of false positives and missed tax frauds. It's possible to complete the work computationally, but the steps are trickier than they seem. Still, value hierarchies, which translate values and norms into design requirements, can facilitate interdisciplinary conversations about integrating ethical standards in AI design.

The suggested AI VSD methodologies shed light on AI-specific value issues throughout system development. AI for SG-VSD requires a multidisciplinary framework. Umbrello and van de Poel don't consider multidisciplinary work. The authors' ignoring these difficulties is troubling, especially given their abstract, highly conceptual prototyping technique.

Despite diverse efforts, they propose anchoring VSD on traditional system development. They misjudge the fact that AI projects require data-oriented architectures, which means that traditional system development methodologies cannot help AI project activities (Khang et al., 2023c).

The main differences arise from unique activities like data discovery, dataset preparation, model training, deployment success measurement etc. Some of these activities cannot be defined precisely enough to have a reliable time estimate, some assume huge potential risks, and some make it difficult to measure the overall added value of the project. Therefore, ML deployments do not lend themselves well to widespread

approaches to software engineering management paradigm, and neither to common software architectural patterns.

This study argues that developing bridges between AI developers and domain experts is vital for establishing robust communication channels and aiding data exploration. Participatory “data-centric AI Ethics by Design” can engage domain experts during system design and enhance corporate understanding of data-driven knowledge generating challenges (Khang et al., 2023d).

10.5 BUILDING BRIDGES IN PARTICIPATORY DATA-CENTRIC AI:
A DESIGN-BASED APPROACH

Ng and Brown (May 9, 2023) focuses on the modern issues in the field of data science and emphasizes the need to -change our thinking toward not only improving the code but toward a more methodical way of improving the data.

In a similar vein, an empirical investigation into the difficulties that data scientists face and the best practices that they employed, they noted that “factors that complicate data understanding include lack of documentation, inconsistent schemas, and multiple possible interpretations of data labels” (Khang, Chowdhury & Sharma (2022).

Initiatives such as the datasheets for datasets that were described earlier establish defined criteria for the documentation of datasets. The process for workforce planning is shown in [Figure 10.4](#).

This may promote transparency and accountability, and it may also enable better communication between dataset makers and dataset consumers. In a manner of speaking, “human-centered approaches” to data science are beginning to acquire traction in the field of human computer interaction (HCI).

For instance, we demonstrate how “data may be foregrounded as an explicit part of design.” It is described how to conduct code-sign exercises that are didactically designed to support group workshops that help subject matter experts can understand and offer insightful criticism on the data and data structures in a particular database system.

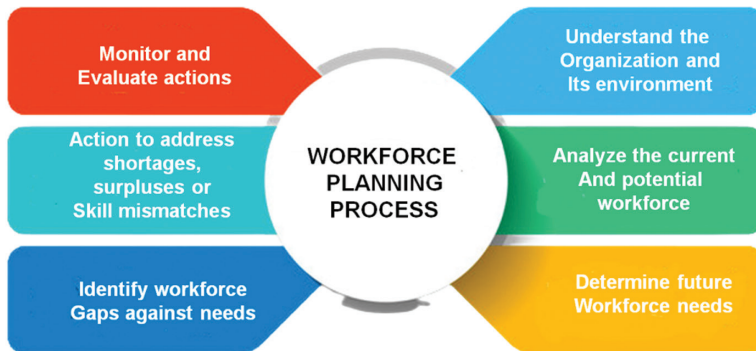


FIGURE 10.4 The process for workforce planning.

Professionals already well-versed in the topic would benefit most from these sessions. Collaboration-based code-sign activities have this effect, making a positive contribution to the empowerment of domain experts and the promotion of data literacy.

Nonetheless, the authors illustrate the complications of database-based services with a real-world example. A data notation, which is made up of straightforward icons that represent the items in a database, it is used by experts in a certain subject to probe into and negotiate the significance of data and the interdependencies between data (Hajimahmud et al., 2022).

The goal of the co-design activities is to make it easier for domain experts to grasp the information architecture of a database and the role that its data entities perform.

To address challenges in a data-driven setting where machine learning is being developed, this contribution is less useful. Choosing the right dataset is crucial to the success of any machine learning project. However, we provide illustrations of how collaborative activities might facilitate domain expert empowerment. Furthermore, they stress the need of putting end users at the forefront of design processes.

Defining the project, gathering and cleaning the data, building and testing the model, and putting it into production are all stages that make up the life cycle of a machine learning project. The recommended approach for AI ethics by design is a participative data-centric design approach, which lays special emphasis on data-related activities and the deployment phase.

Concerns about model training and verification are not addressed, though. One of the motivating factors is the fact that ML engineers spend up to 80% of their time on data preparation. There is a discrepancy between the unbiased nature of ML research and the realities of ML development practice, according to an investigation into the requirements of actual machine learning practitioners.

The authors discuss how the impact of the dataset is overlooked in favor of the development of “algorithmic de-biasing methods” in the research literature. But they observe that “many of our interviewees highlighted that their teams often look to their training datasets, rather than their ML models, as the most critical spot to intervene to promote fairness in their products.”

Because of this, it is essential to first have a deep grasp of the data in a particular domain in order to confidently carry out the next processes, model learning and verification.

In light of these findings, machine learning engineers occasionally face the issue of evaluating and comprehending data complexity without the assistance of domain expertise. The need to communicate whether or not domain-specific data reflects trustworthy and verified representations of a given practice is also not always understood by end users.

Since it is the intended audience who will be putting the data to use, this is a concern. This difficulty is less of an issue in logically organized topics than in those with less formalized procedures. A medical image analysis system, for instance, relies heavily on human labeling of training data to ensure accurate results in detecting breast cancer.

Further, the quality of the labeling effort determines the predicted accuracy of the machine learning system. If it turns out to be challenging to obtain professionals to carry out the labeling process, this could become a bottleneck issue for the machine learning project.

In a broader sense, however, the health domain is understood to be a highly theoretically structured industry with clearly delineated labor patterns and objective diagnostic standards. This paves the way for efficiently separating out domain-specific truths.

However, in other domains, like public administration, where the data represent socially created phenomena, such as social care or education, effective data labeling may be more challenging. To figure out, for instance, the term “educational excellence” needs to be defined with the help of specialists in the field to ensure it accurately reflects what is needed for students to achieve their goals.

To make a well-informed choice on which proxies to use, these specialists also need an understanding of the processes underlying the creation of knowledge using data. For similar reasons, statistics may include validated explanations of concepts like “disadvantaged individuals” without first considering the possibility of ambiguity in such representations. Algorithms that are trained on statistics from such validated data may make inaccurate predictions about vulnerable people, leading to unwarranted interventions.

Consequently, “the issue of picking what may be quantified in order to offer good predictions should not be overlooked.” Our recommendation for systematically improving the data activities is to engage in participatory activities while consulting with experts in the sector. Historically, the Scandinavian heritage of system development has served as an inspiration for the participatory design techniques used in HCI.

Users were considered “capable practitioners” and actively participated in the creation of ICT systems for use in the workplace under this tradition. Through the process of participatory design, a channel of communication is opened up between those working on the system’s development and those who will ultimately be utilizing it.

By conceptualizing the design process as a process of developing new language games that have a family resemblance with the language games of both users and designers, we have an orientation for practicing design as skill-based involvement, which written by Snehal, Babasaheb and Khang (2023).

The idea of language as games put out by Wittgenstein inspired them to make this observation. The performance of a machine learning model is typically not dependent on the modeling technique but rather on the data preparation. If the person performing this data preparation understands the subject or has access to appropriate domain expertise, then the work will be of sufficient quality.

Since “a model is not better than the predictor variables inputted to it,” and “domain expertise allows the derivation of powerful predictor variables from the current variables,” it is clear that specialized knowledge is indispensable. In this light, it becomes clear how crucial it is to ensure that all audience members have a firm grasp of the presented data domain. Thus, by supplying dialogical guidelines for developing competence in a particular topic, “bridge builder” may host “a conference of language games,” promote collaborative design projects, and support participatory design activities.

To expose the domain knowledge against which the data must be comprehended, the bridge builder might conduct collaborative workshops and introduce tools like conceptual sketching, thought mapping, and card sorting. With this, the bridge builder can make way for mutual understanding (Khang et al., 2022a).

There is a good chance that the domain expert is feeling disoriented and out of his or her depth when it comes to ML development. In this case, starting with an investigation into the domain expert's competence could help the domain expert feel more confident. From the perspective of the programmer, learning about the knowledge the domain expert possesses would improve his or her chances of spotting nuances in data later on.

The domain might be better represented in the datasets provided if he or she has this kind of background knowledge, or the datasets might be incorrect because the people providing them don't realize how crucial it is to deliver accurate and validated information. Expert subject knowledge clarification also helps find common ground before diving into the more difficult work of analyzing value systems and power dynamics.

This chapter's second section discusses value elicitation phases in the context of AI VSD design techniques; therefore, it seems to make sense that these would coincide with the phases during which values are investigated as being at risk. Value hierarchies and value drawings are two examples of techniques that can help get the ball rolling on value elicitation activities. Because of this, we get an opportunity to reflect on our own confusion over the data's value problems (Khang, Rani & Vrushank, 2023).

Additionally, the use of agile consequential scanning for responsible innovation and semi-structured value interviews can support value elicitation activities and serve as a starting point for discussions about the epistemological and ethical issues associated with data-driven knowledge generation in the area of study.

While the above-mentioned AI VSD approaches recognize the existence of societal bias and propose tactics for de-biasing, they do not tackle the larger power structures that surround a data environment; instead, they suggest changes in organizational workflows as a possible fix.

Understanding the manner in which racially and gender-based picture databases reflect and sustain institutional discrimination can be undertaken with a considerable level of simplicity. We also stress the importance of the bridge builder's responsibility to educate ML experts on the influence of power dynamics on data's underlying truth.

To accomplish this, it is necessary to increase understanding of the role that power dynamics can play in distorting data's foundational truth. The ways in which power structures might influence data in an organizational setting are typically relatively subtle because of the complex nature of the interplay between social and organizational institutions.

For instance, the tacit dimension of the care practice that develops is not sufficiently captured by the data used as the foundation for DM support systems in the senior care industry, undervaluing that which cannot be assessed. This "lends itself to an instrumental approach that is not supportive of developing skills within the field of care giving," and the "datafication" of the senior care business, as a result, reflects a political desire for documentation and standardization.

The need for documenting also has a negative effect on careers' overall workload, which might lead to less-than-reliable data. These people view these pursuits as bureaucratic overhead that distracts them from their real jobs.

Petersen, Christensen, and Hildebrandt show in a recent study that social case-workers are hesitant to submit sensitive data about clients into AI systems, making it challenging for algorithmic decision-making in public administration.

This complicates the use of algorithmic decision-making in the public sector. It creates ethical challenges because the formal record is decontextualized and does not reveal information about the real-world framing of situated decision-making processes.

As an added bonus, they worry that consumers' future opportunities may be shaped and damaged by sensitive data. Even Eubanks, who warns that "equity needs the freedom to develop, whereas the digital poorhouse confines us into patterns of the past," is concerned about this. With his remark, Eubanks repeats past concerns.

There are no quick fixes or easy strategies for resolving the political contradiction between datafication and practice. However, the role of the bridge builder is to highlight these challenges so that ML developers can learn to better foresee the wider repercussions of their work. More broadly, we need to cultivate a practice that makes ethics as natural as possible by promoting the moral self-cultivation of the community of ML developers through educational programs.

As a corollary, we should be mindful that engineering activism requires an understanding of the interconnectedness of the world's power systems and its most pressing challenges. The goals of most ML projects are designed to maximize money at the expense of people's genuine desire for a sustainable future.

A move to "revitalize participation by changing such that it may once again become a tool to assist people impact serious problems in their lives" seems warranted at this time. This offer is not meant to get the HCI community involved in "the Competition State," but rather to get them involved in designing a more sustainable way of life. This ambitious goal is reflected in the AI4SG-VSD design approach, in one form or another (Sienna, 2021).

All of the AI VSD approaches discussed so far agree that, at the end of the deployment life cycle, it is crucial to keep a close eye on how well AI systems are doing by using impact assessment tools and thinking carefully about whether and how the introduction of AI affects workflow and organizational structures. It's as crucial to pay attention to that as it is to the other problems, since the concealed technical debt in ML systems creates continuous maintenance difficulties.

Cunningham (1992) used the phrase "hidden technical debt" as an analogy to budgetary debt in the deployment of conventional software systems to emphasize how a balanced technological debt may be both inevitable and beneficial, but may also present significant challenges if not controlled. This was done to emphasize the need to keep in mind that technical debts may be harmful if they are ignored, even though they might be useful in some circumstances.

Maintainable machine learning is still in its infancy, which is made even more difficult by the challenge of hidden technical data debt. ML systems are more susceptible to configuration debt than regular software systems because ML infrastructures are so complex and require ML-specific configurations in addition to the configurations that are typically used to create conventional software systems.

In addition to technical debt at the code level, ML systems must also consider technical debt at the system level. This is especially crucial in the case of external data dependency debt, which can build up undetected due to the subtle nature of feedback loops between systems that only exert a latent influence on one another.

Additionally, as a result of the updated model, people are now more likely to get information on the flu. In addition, while ML may function effectively, there are no

free meals, as shown in the following citation: Mature systems may have dozens or hundreds of models running simultaneously this raises a wide range of important problems, including the problem of updating many configurations for many similar models safely and automatically, how to manage and assign resources among models with different business priorities, and how to visualize and detect blockages in the flow of data in a production pipeline.

Therefore, DM should be made aware of the type of organizational commitment needed to properly manage the burden of dedicating resources to taming and paying off technological debt in such systems from the very beginning if they are considering the adoption of ML systems (Tailor, Pareek & Khang, 2022).

10.6 CONCLUSION

The goal of this chapter is to provide a data-centric, participatory approach to AI ethics during the design phase. The author suggests learning from modern AI VSD methods as a starting point. In this setting, a person with expertise in both computer science and ethics is needed to help bring together the many parties involved in order to reach a consensus.

By examining and integrating the concepts of changing and organizational flexibility into a conceptual data framework, this research makes a significant contribution to the field of change management. Using a real-world case study supplied by our industry partners, we constructed a relational data model and a physical database system to demonstrate the efficacy of our meta-model (Misra et al., 2023).

Findings suggest that our model can be utilized to help with the decision-making involved in the change management process in real-world organizations. Therefore, starting from a data-centric approach allows for ML developers and domain specialists to work together on projects revolving around a certain data domain and its relationship to a specific practice.

Because the data-centric viewpoint is used as a starting point, this is achievable. Such a data-centric setting provides a concrete scenario for more theoretical considerations of value in the context of designing and deploying machine learning systems.

The end result of this effort is the creation of ML-based systems. Therefore, tasks involving the shared collecting of data are a realistic place to begin applying AI ethics by design. Bridge construction is an important part of establishing reliable channels of communication and cataloguing different types of domain-specific knowledge in ML projects.

This chapter has described one approach to AI ethics by design that can make a reasonable contribution toward that goal: A participative data-centric design approach (Khang et al., 2023d).

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11 Prediction of Employees' Performance Using Machine Learning (ML) Techniques

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11.1 INTRODUCTION

Employees are the key assets of every firm. Every business requires adequately trained workers to accomplish their duties successfully and efficiently (Karwehl & Kauffeld, 2021). Human resources are highly important capital in industries and organizations; they are people with education, knowledge, experience, skills, and competence to work in enterprises. Therefore, the competitiveness of the organization depends on the knowledge and skills held by its personnel (Khang et al., 2023b).

The quality of human personnel is one of the criteria required to increase organizational productivity. Therefore, an organization has to evaluate the performance of its staff (Midhat Ali et al., 2021). Management of human resources in enterprises plays a significant part in enterprise growth, and it is a vital support for fighting the risk of market competitiveness.

Human resources become a very crucial feature for an organization's connection in producing value, flexibility, accomplishment, and developing its competency and flexibility (Semenova, Zelenyuk & Savinov, 2021).

It's not always enough for employees to perform the task to obtain a competitive edge, and this is where creativity and innovative behavior come into play (Sedyastuti et al., 2021). Companies increasingly look for personnel that can think creatively and innovatively in this digital and global era. It's difficult for companies with a lack of creativity and innovation to succeed in the marketplace.

For a company to stay competitive in the marketplace, it is essential to foster the creativity of its employees. Additionally, a variety of businesses are always on the lookout for new approaches to encourage their workers to work with originality and creativity (Widarni & Wilantari, 2021). There is a lot of interest in understanding how employee creativity, capacities, and skills change over time to identify the elements that promote or impede a firm development.

Performance evaluation or appraisal is one of the most important techniques of human resource management for assessing the efficiency of workers in the workplace (Choudhury, 2021). Employees are more likely to work efficiently if they are given a thorough review of their performance.

Performance assessment theory is extensively used in human resource management, and corporations use it to monitor, evaluate, and enhance the company's economic efficiency and organizational management. Several companies recognize that the performance evaluation process is tedious (Dobrowolski, Drozdowski & Ledzianowski, 2021). Traditional methods of evaluating employee performance rely on a single criterion, such as the amount of work completed.

Multi-criteria decision-making is necessary for assessing the performance of workers due to many competing criteria (Drozdowski, Rogozińska-Mitrut & Stasiak, 2021). Despite the great advances in technology, most firms still use manual performance evaluations.

The manual employee evaluation methods are susceptible to human bias, since supervisors may evaluate the performance of their staff personally (Snehal, Babasaheb & Khang, 2023a).

The existing human resources data system does not accurately anticipate how productive a person will be in the future for the organization (Islam & Amin, 2021). Conventional business models are outmoded and do not help firms to make wise judgments. The common biases in traditional employee performance evaluation systems are the halo effect, horn effect, leniency bias, strictness bias, recency bias, personal bias, and contrast bias.

Figure 11.1 illustrates the explanation of these biases (Mehta & Ali, 2021). When managers aren't given enough training to provide objective feedback, biases might creep in. Traditional analytic approaches are rendered ineffective by their reliance on a small number of assessment criteria and their high computing costs (Dencik & Stevens, 2021). Furthermore, managers don't prioritize performance reviews since they're a time-consuming activity.

All of these factors contribute to a skewed evaluation of performance, which leads to dissatisfaction among workers who receive it.

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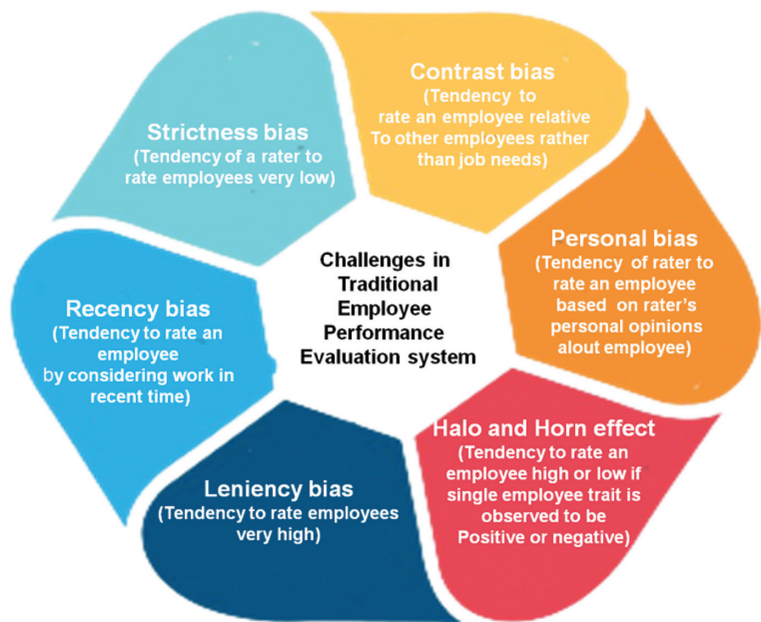


FIGURE 11.1 Biases in the traditional employee performance evaluation system.

2021). Today, companies are steadily exploring the performance evaluation model for increasing productivity. (Douglas, 2021) Scientific investigation of the performance assessment system is required to further improve the economic efficiency of businesses as a whole.

As a result, the focus of this chapter is on the development of a machine learning (ML)-based system for evaluating employee performance, which can be used by companies to reduce assessment bias and help workers and management better understand how well they are doing and where they may improve (Mwaro, Ogada & Cheruiyot, 2021).

11.2 PROBLEM STATEMENT

Employee efficiency is widely recognized as critical to an organization’s overall production and success. Multi-objective decision-making must be used when evaluating human resources performance in the workplace rather than merely counting the number of activities performed. An organization’s total success is strongly influenced by its ability to evaluate its employee performance (Khanh & Khang, 2021).

If an organization is serious about increasing its market share, it requires an effective performance evaluation system that incorporates a wide range of criteria for management employees. The use of ML methods in the assessment of employee performance is increasing. This creates motivation for developing an ML-based employee performance evaluation model (Rana et al., 2021).

Here, a support vector-based employee performance evaluation model is constructed. A traditional support vector machine (SVM) is mainly used to classify the data into two categories, and its accuracy must be improved. To solve these issues, we proposed a weighted multilayer support vector machine (WM-SVM) classifier (Rani et al., 2023).

11.3 PROPOSED WORK

The main objective of this chapter is to develop a WM-SVM-based employee performance evaluation model for classifying employees into three categories: namely, high, moderate, and low performance (Bhambri et al., 2022).

Figure 11.2 depicts the framework of the proposed work. Initially, employee personal and professional competency details were collected and processed to ensure the quality of data. Relevant features for evaluating employee performance were extracted using principal component analysis (PCA).

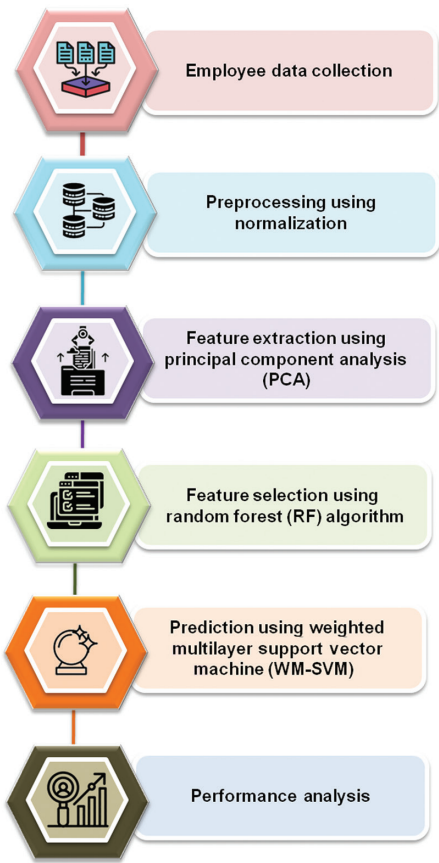


FIGURE 11.2 Framework of the proposed work.

Most significant employee performance evaluation features were selected using random forest (RF) to build an optimal feature set for employee classification.

Finally, employees were classified into different performance categories based on the optimal feature set using WM-SVM as shown in [Figure 11.2](#).

11.3.1 EMPLOYEE DATA COLLECTION

The sample data for this study was collected respective to 100 employees of a medium-sized firm in the Xi’an high-tech industrial development zone in Shaanxi Province, China. Their demographic data were collected initially (Gupta et al., 2023).

Questionnaires are framed based on multiple performance attributes, such as knowledge (educational level, professional knowledge), work performance (quantity of work, work quality, time management), leadership and communication (team-work, cooperation, leadership skills, and communication skills), and intrapersonnel skills (problem-solving skills, innovation, and creativity, confidence, behavioral attitude, flexibility) (Babasaheb et al., 2023a).

Questionnaires were distributed to human resource evaluation managers at the firm to provide their views and ratings for each question to evaluate the performance of an employee (Babasaheb, Sphurti & Khang, 2023b). For each employee, feedback from various human resource evaluation managers of the firm is collected as shown in [Table 11.1](#).

TABLE 11.1
Demographic Characteristics of Sample Employees

	Variables	Frequency	Percentage (%)
Age	18–30	20	20
	31–49	45	45
	50–65	35	35
Gender	Male	60	60
	Female	40	40
Marital Status	Married	31	31
	Unmarried	45	45
	Divorced/widow	24	24
Educational Level	High school	17	17
	Diploma	23	23
	Bachelor Degree	60	60
Experience at firm	1–5 years	34	34
	6–10 years	25	25
	11–15 years	31	31
	Above 16 years	10	10
Employee Position	Manager	21	21
	Marketing officers	45	45
	Operational officers	25	25
	Entry-level employees	9	9

11.3.2 NORMALIZATION OF EMPLOYEE EVALUATION DATA

The employee evaluation data was preprocessed to scale the ratings provided by human resource evaluation managers for an employee, within a particular range of -1 to 1 . The employee evaluation data was transformed by moving the decimal points of values for the performance evaluation attribute (Khang et al., 2023a).

The number of decimal points moved depends on the maximum absolute value of the performance evaluation attribute. A value of employee evaluation data is normalized using equation 11.1.

$$r' = \frac{r}{10^x} \quad (11.1)$$

where r' means the normalized value of employee evaluation data, r denotes the original value of employee evaluation data, and x represents the smallest integer such that $\max r' < 1$.

11.3.3 EMPLOYEE EVALUATION FEATURING EXTRACTION USING PRINCIPAL COMPONENT ANALYSIS (PCA)

Relevant features for evaluating employee performance were extracted using PCA. Transforming high-dimensional employee evaluation data into a lower-dimension PCA reduces complexity without losing important trends or patterns (Jain et al., 2023).

Assume $[E]$ as training data composed of employee evaluation data with k dimensions. The input matrix to PCA is defined by equation 11.2.

$$[E] = \{[E_1], [E_2], \dots, [E_n]\} \quad (11.2)$$

Equation 11.2 can be expanded into equation 11.3.

$$E_{n \times k} = \{(e_1^1, e_2^1, \dots, e_k^1), (e_1^2, e_2^2, \dots, e_k^2), \dots, (e_1^n, e_2^n, \dots, e_k^n)\} \quad (11.3)$$

where e_k^n means the k th evaluation data of the n th employee. The average evaluation data of each employee ($\overline{E_n}$) is calculated by equation 11.4.

$$\overline{E_n} = \frac{1}{k} \sum_{j=1}^k e_j^n \quad (11.4)$$

where k denotes the number of employee observation data and e_j means the employee observation data.

The covariance matrix is determined using equation 11.5.

$$C_n = \frac{1}{k-1} \sum_{j=1}^k (e_j^n - \overline{E_n})(e_j^n - \overline{E_n})^p \quad (11.5)$$

where C_n refers to the covariance matrix, k denotes the number of employee observation data, e_j means the employee observation data, and $\bar{E}_n$ is the average of evaluation data of each employee.

Then the eigenvectors and eigenvalues of the covariance matrix are determined. To reduce the dimensions of employee evaluation features, the cumulative proportion of variance is used.

The cumulative proportion of variance (CPV) is defined as the eigenvalue for a specific evaluation feature out of the total eigenvalues defined for the training data matrix [E] and is calculated using equation 11.6.

The CPV for each employee evaluation feature is defined using equation 11.6. This is done to normalize the eigenvalues determined for evaluation features.

$$CPV_j = \frac{\lambda_j}{\sum_{m=1}^n \sum_{j=1}^k \lambda_j^m} \times 100 \quad (11.6)$$

where CPV_j denotes the CPV of the j th evaluation feature, λ_j represents the eigenvalue of the j th evaluation feature, and $\sum_{m=1}^n \sum_{j=1}^k \lambda_j^m$ denotes the total eigenvalues defined for the training data matrix.

The CPV for each employee evaluation feature was compared to the threshold value. The threshold is a constant and set as 85%. The employee evaluation features that satisfy the criteria (CPV of employee evaluation feature must be higher than the threshold value) provided in equation 11.7 were extracted.

$$\frac{\lambda_j}{\sum_{m=1}^n \sum_{j=1}^k \lambda_j^m} \times 100\% > Threshold \quad (11.7)$$

The dimensionally reduced employee evaluation data matrix is constructed by projecting the original employee evaluation data matrix according to the extracted evaluation feature set. It is

$$E'_{n \times l} = E_{n \times k} \times F \quad (11.8)$$

where $E'_{n \times l}$ means the dimensionally reduced employee evaluation data, F refers to the extracted evaluation feature set. Thus k dimensional employee evaluation data is reduced to l dimensional employee evaluation data using PCA.

11.3.4 EMPLOYEE EVALUATION FEATURE SELECTION USING RANDOM FOREST (RF)

Significant features for an employee performance evaluation are selected from the extracted evaluation feature set, obtained from the above step, using RF. RF is an ML method depending on the decision tree (DT), which has great generalization power owing to two built-in randomization mechanisms in the algorithm.

The first way for introducing randomization involves drawing training samples for employee evaluations at random from the whole pool of employee assessment results using bootstrap resampling methods.

The second kind of randomization occurs during tree development, when subsets of evaluation characteristics are chosen at random from the whole collection of features. The steps involved in the computation of feature importance for an employee evaluation feature are illustrated in [Figure 11.3](#).

Step 1: ‘ v ’ training employee evaluation datasets are randomly extracted from original employee evaluation data using the bootstrap method. The unselected employee evaluation data are randomly split into ‘ v ’ testing employee evaluation datasets which are also known as out-of-bag (OOB) sample sets. The training employee evaluation datasets are termed as $TS_1, TS_2, \dots$ and TS_v , and testing OOB employee evaluation datasets are termed as $O_1, O_2, \dots, O_v$.

Step 2: The employee evaluation feature subsets are randomly selected from the entire employee evaluation feature set.

Step 3: When deciding how to divide a node in a tree, only the most relevant attributes from the chosen subset are used. For each employee evaluation training dataset, the decision tree is constructed. The decision tree constructed for ‘ v ’ employee evaluation training datasets is denoted as $DT_1, DT_2, \dots, DT_v$.

Step 4: The value of the employee evaluation feature (e_j) in all OOB employee evaluation testing datasets is permuted into e'_j . The values of other employee evaluation features remain unchanged. New OOB employee evaluation testing datasets are constructed with permuted employee evaluation features and unchanged employee evaluation features. Newly constructed OOB employee evaluation testing datasets are termed as $O'_1, O'_2, \dots, O'_v$.

Step 5: The constructed decision tree is tested on original OOB employee evaluation testing datasets. The error rate in employee performance classification for this dataset using the respective trained decision tree is calculated and denoted as $ER_1, ER_2, \dots, ER_v$.

Step 6: The constructed decision tree is also tested on new OOB employee evaluation testing datasets. The error rate in employee performance classification for these new datasets using the respective trained decision tree is calculated and denoted as $ER'_1, ER'_2, \dots, ER'_v$.

Step 7: For each OOB employee evaluation testing dataset, variations in classification error rate are determined to obtain the classification error rate difference, according to equation 11.9.

$$ER''_j = ER_j - ER'_j \quad (11.9)$$

where ER''_j means the classification error rate difference for the j th OOB testing dataset, ER_j means the classification error rate for the j th OOB testing dataset, ER'_j means the classification error rate for j th new OOB testing dataset, and $j = 1, 2, \dots, v$.

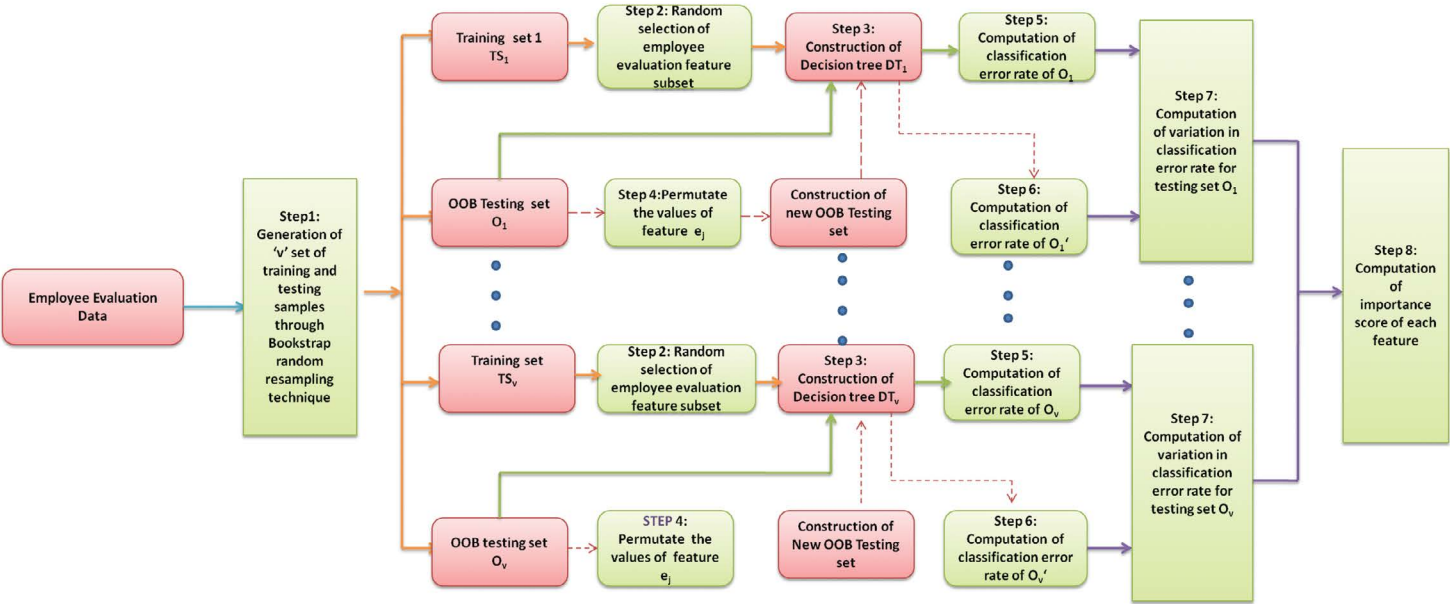


FIGURE 11.3 Flowchart of RF-based employee evaluation feature importance computation.